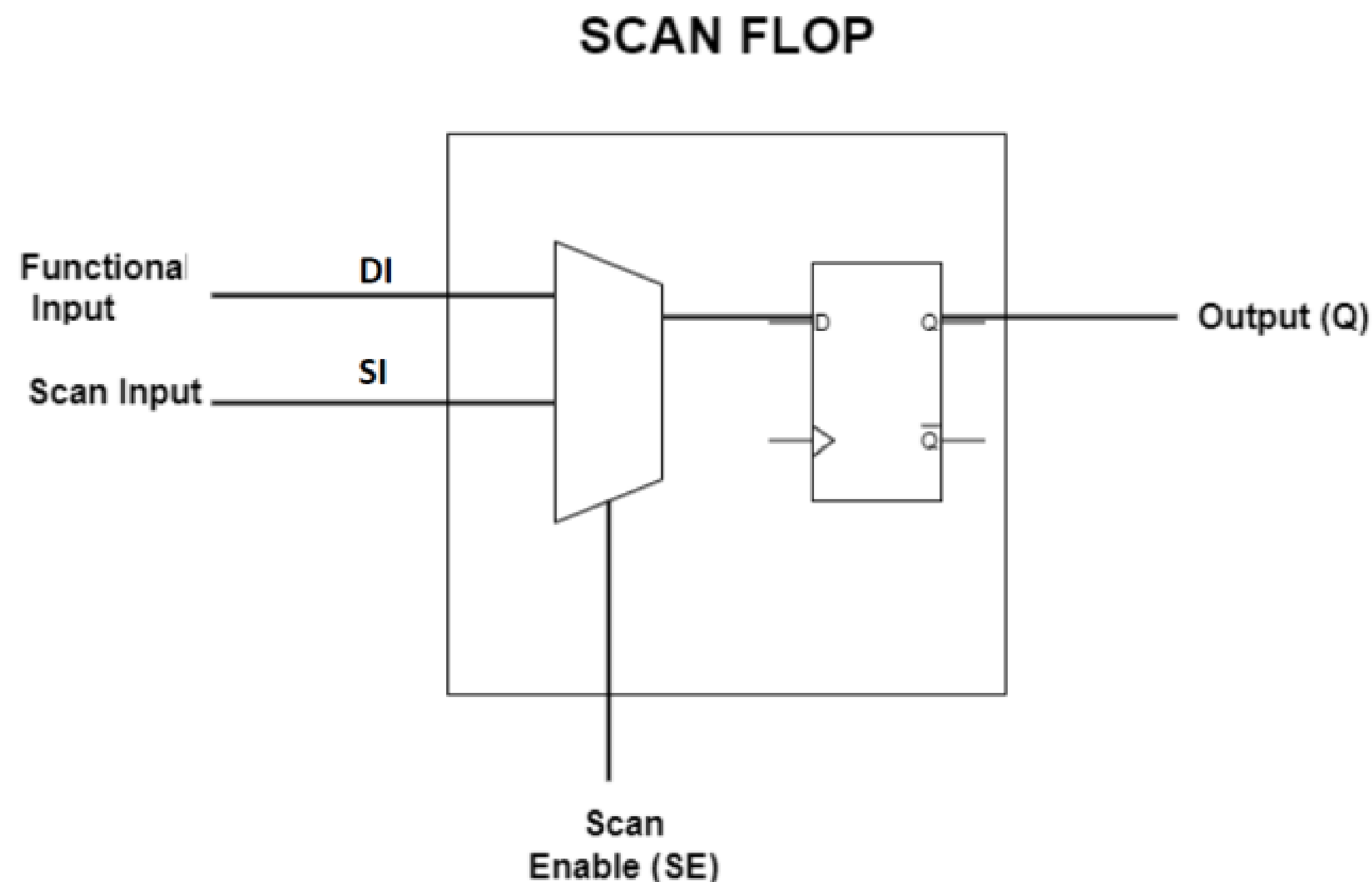


TYPES OF SCAN CELL DESIGNS

Three types :

- Muxed-D Scan Cell
- Clocked-Scan Cell
- LSSD Scan Cell

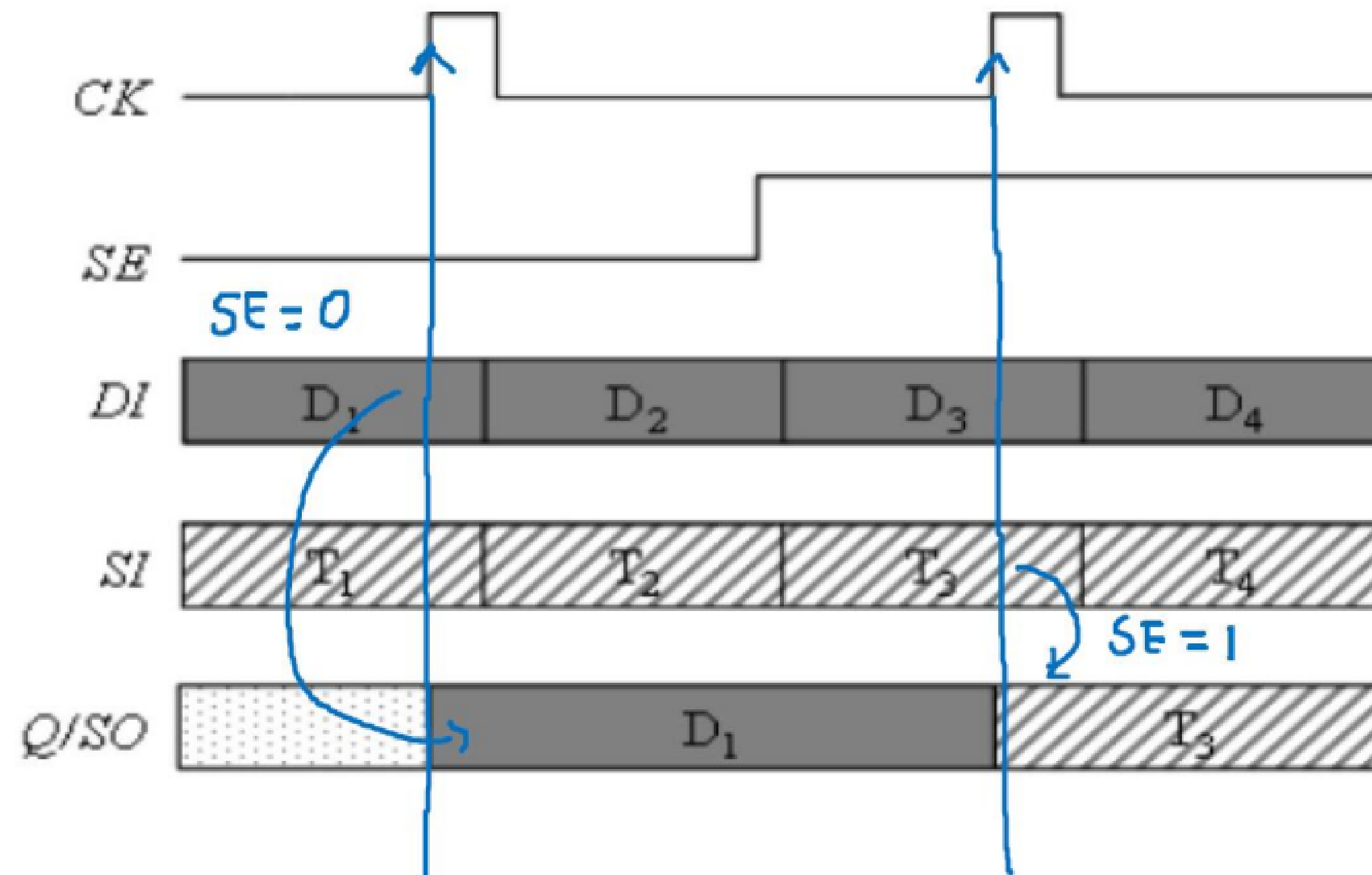
MUXED-D SCAN CELL



This scan cell is composed of a D flip-flop and a multiplexer.

The multiplexer uses an additional scan enable input SE to select between the data input DI and the scan input SI.

MUXED-D SCAN CELL Contd...

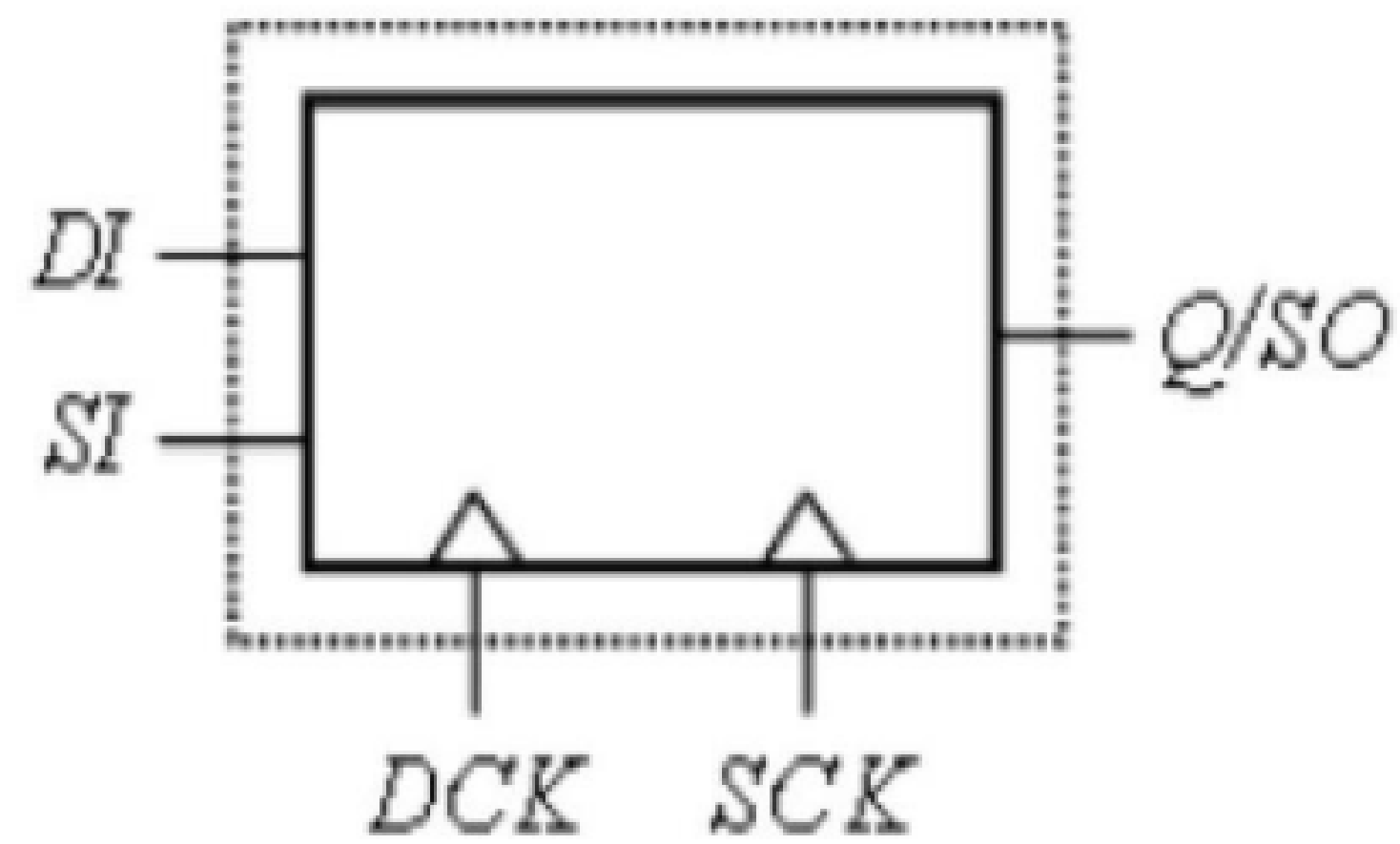


Edge-triggered muxed-D scan cell design and operation

In normal/capture mode, **SE** is set to **0**. The value present at the data input **DI** is captured into the internal D flip-flop when a rising clock edge is applied.

In shift mode, **SE** is set to **1**. The scan input **SI** is used to shift in new data to the D flip-flop, while the content of the D flip-flop is being shifted out.

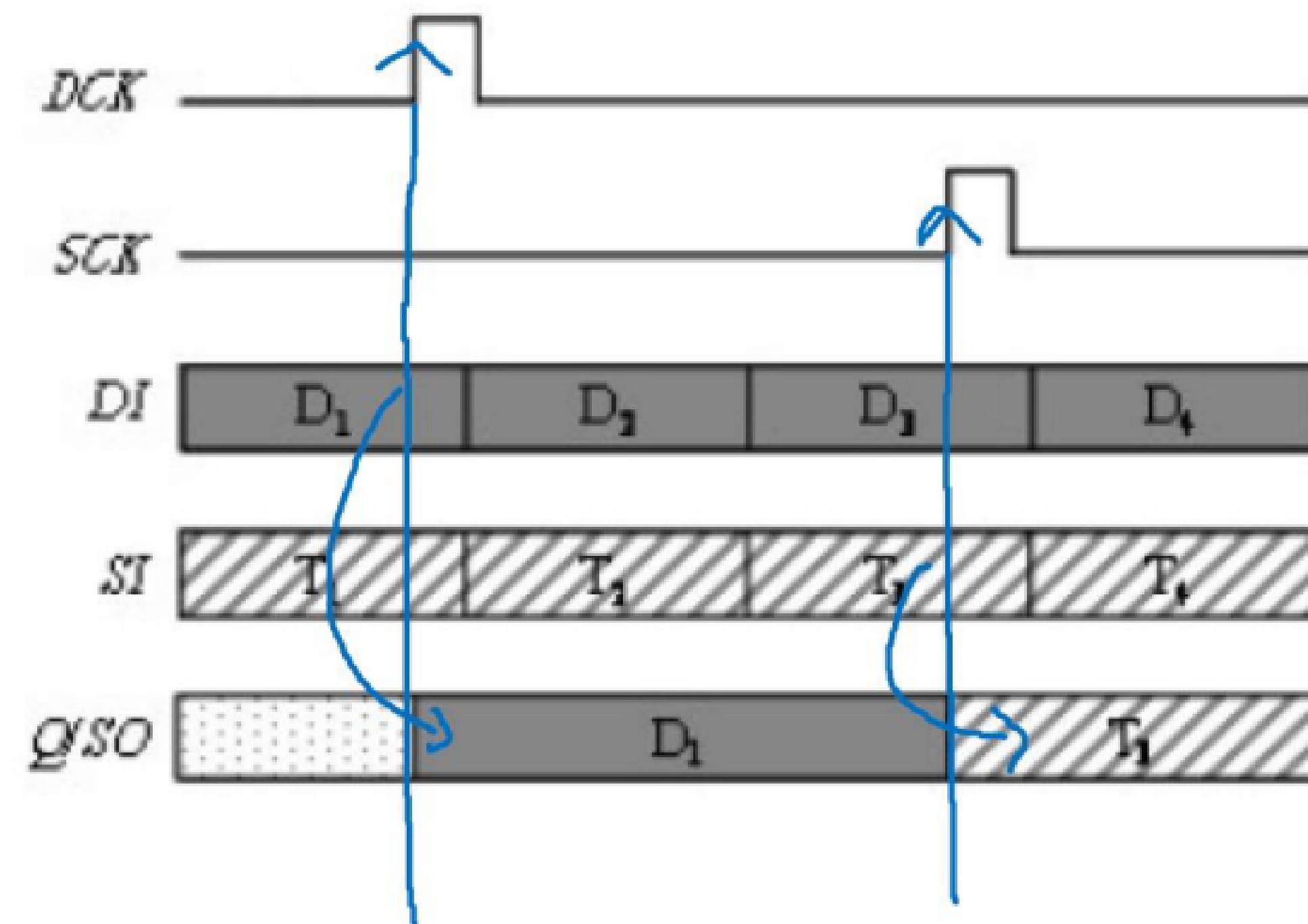
CLOCKED-SCAN CELL



In the clocked-scan cell, input selection is conducted using two independent clocks, *DCK* and *SCK*.

Clocked-scan cell

CLOCKED-SCAN CELL Contd...

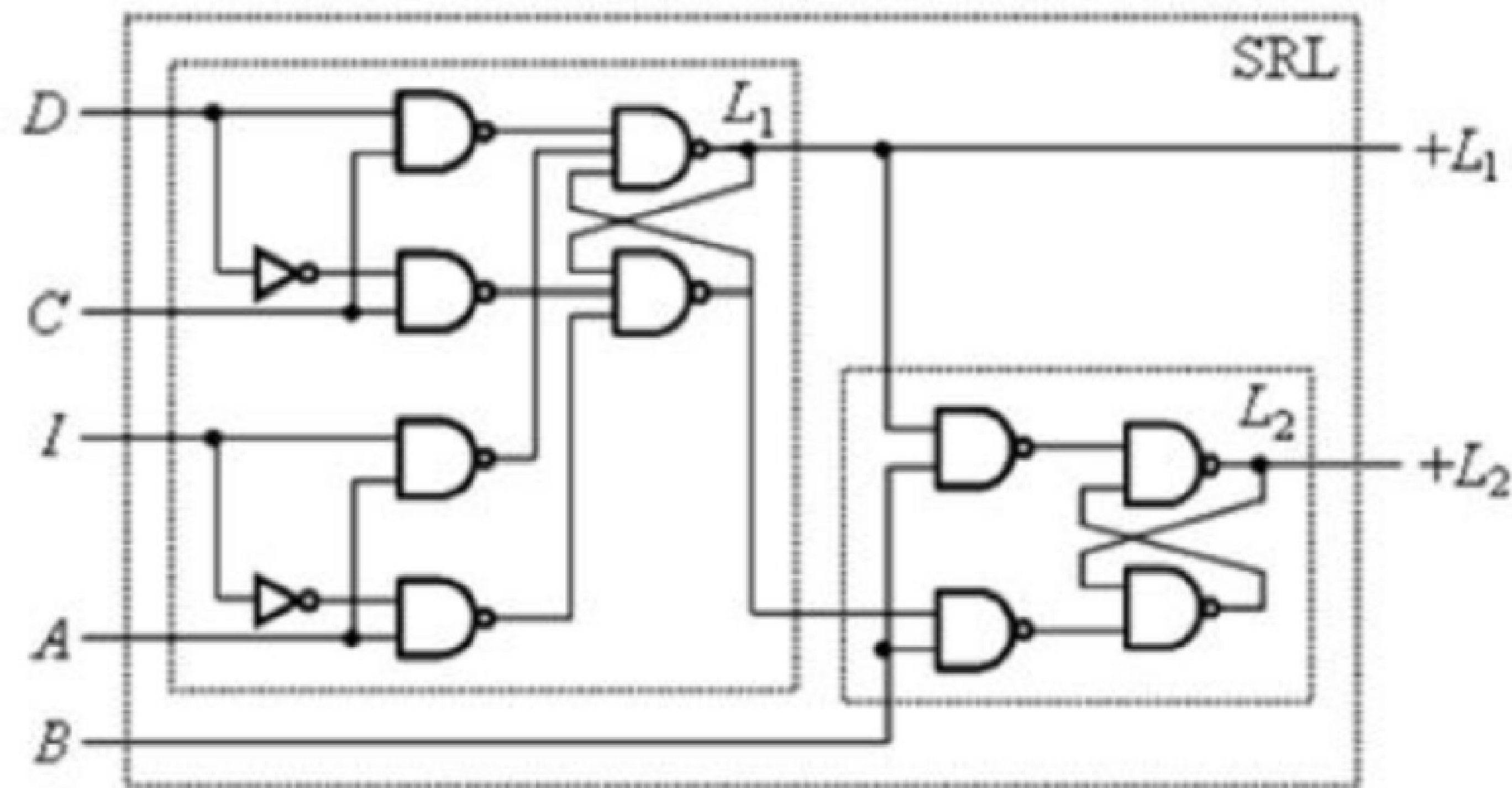


Clocked-scan cell design and operation

In normal/capture mode, the data clock **DCK** is used to capture the contents present at the data input **DI** into the clocked-scan cell.

In shift mode, the shift clock **SCK** is used to shift in new data from the scan input **SI** into the clocked-scan cell, while the content of the clocked-scan cell is being shifted out.

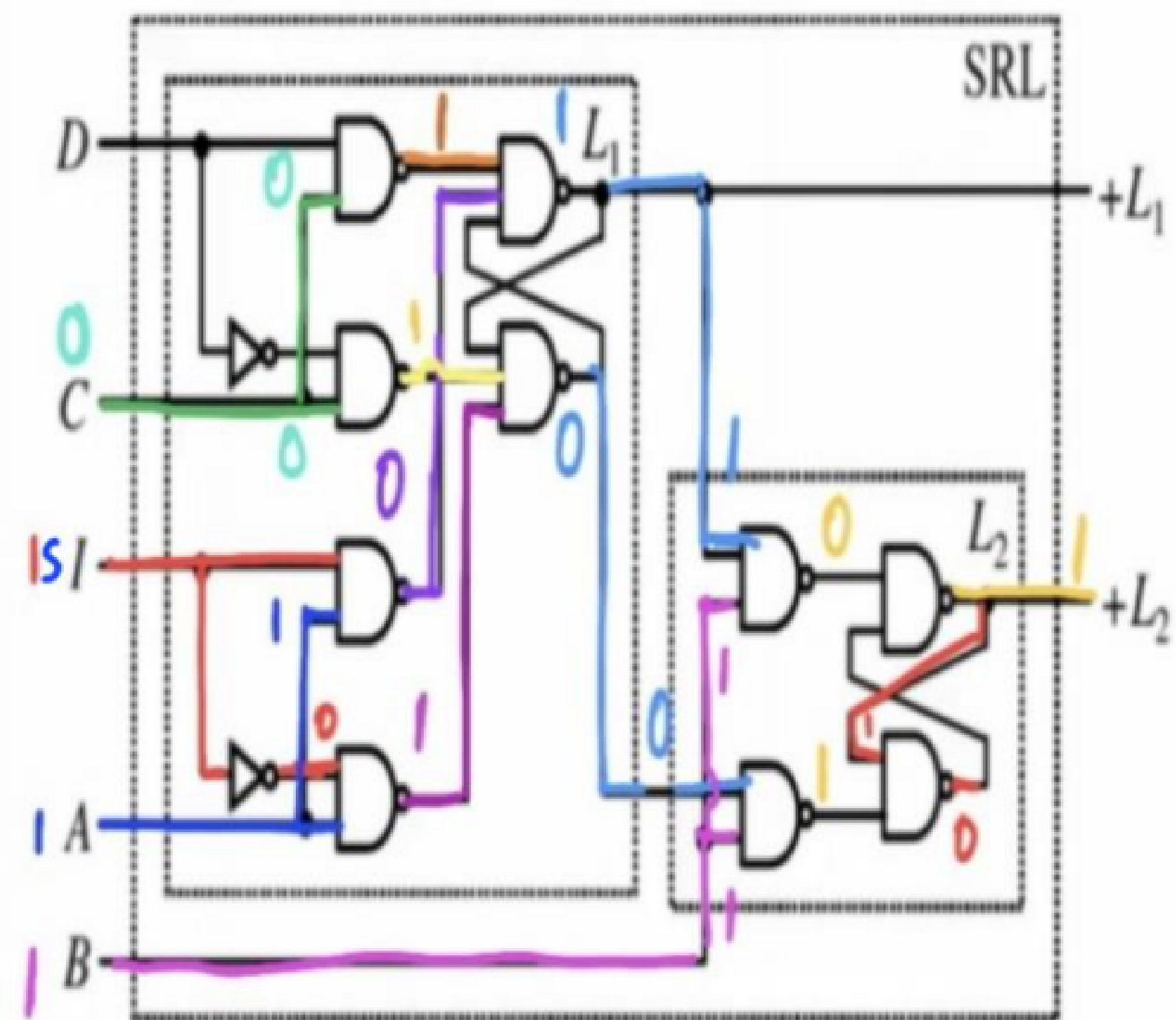
LSSD (Level Sensitive Scan Design) SCAN CELL



Master Latch L1 (A, C clocks)
Slave Latch L2 (B clock)
2 inputs : Data input (D), Scan
Input (SI)
2 outputs : +L1, +L2

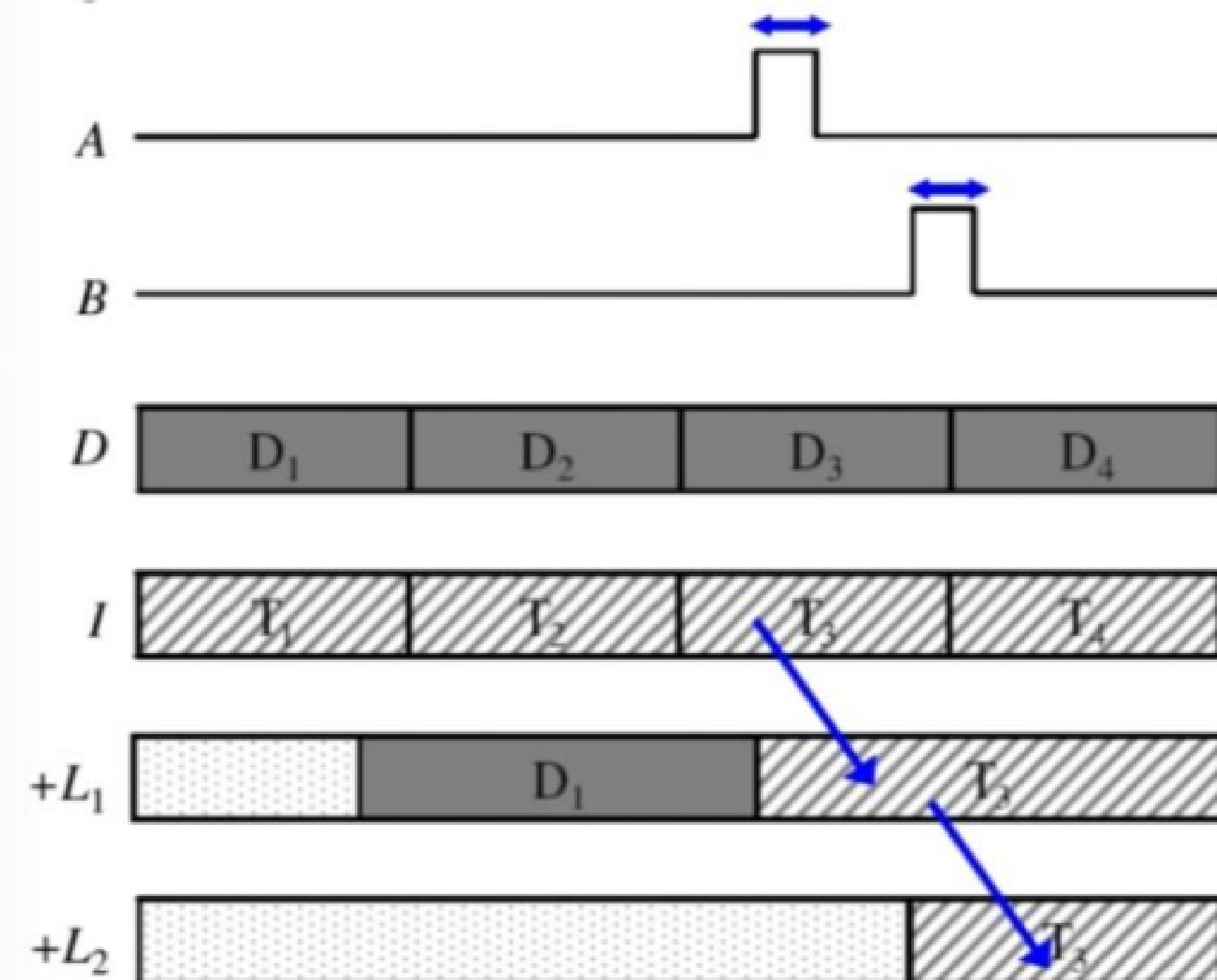
- **Clocking sequence:**

- ♦ **A clock B clock $SI \rightarrow +L_1 \rightarrow +L_2$**



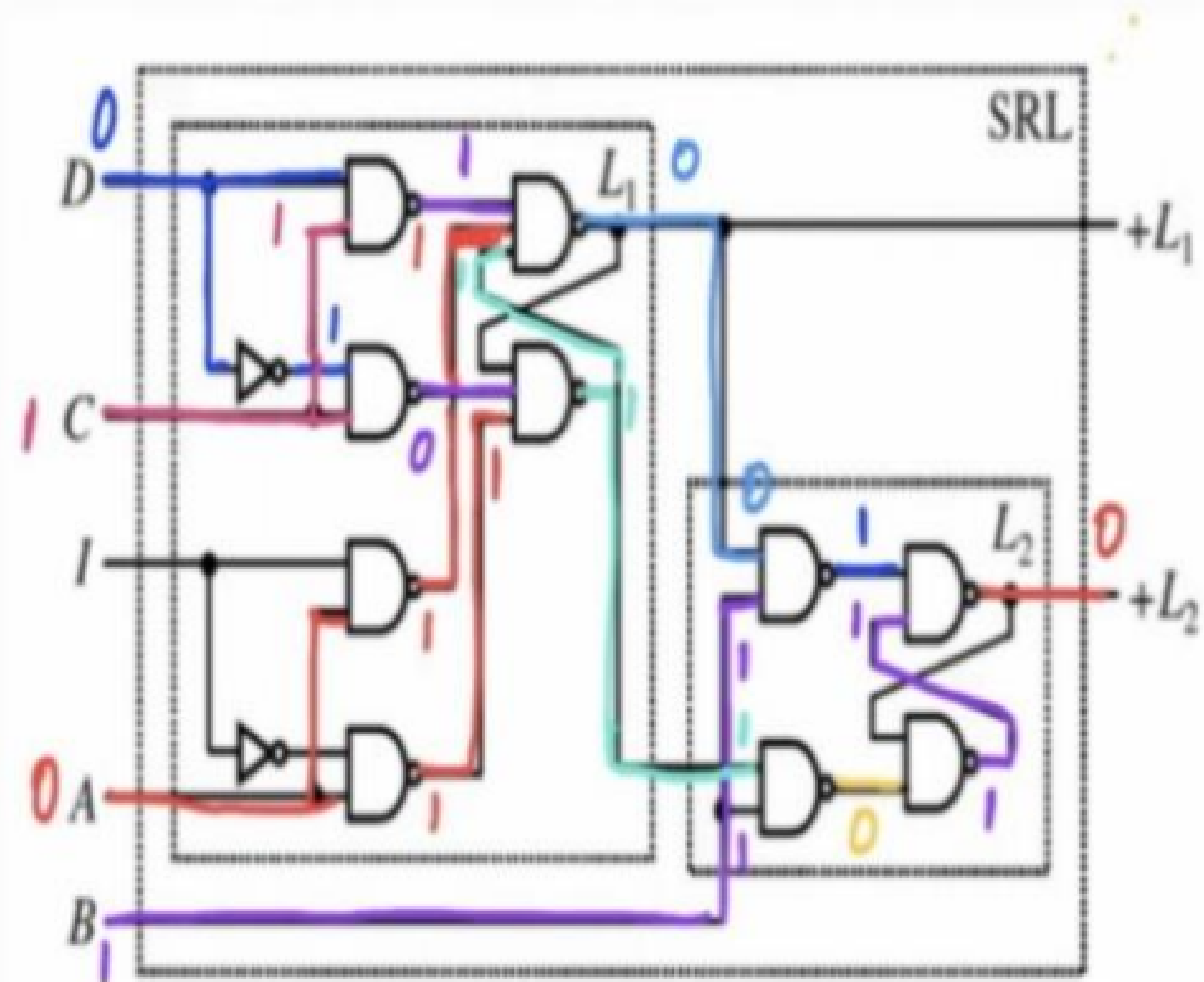
NAND TT

A	B	y
0	0	1
0	1	1
1	0	1
1	1	0



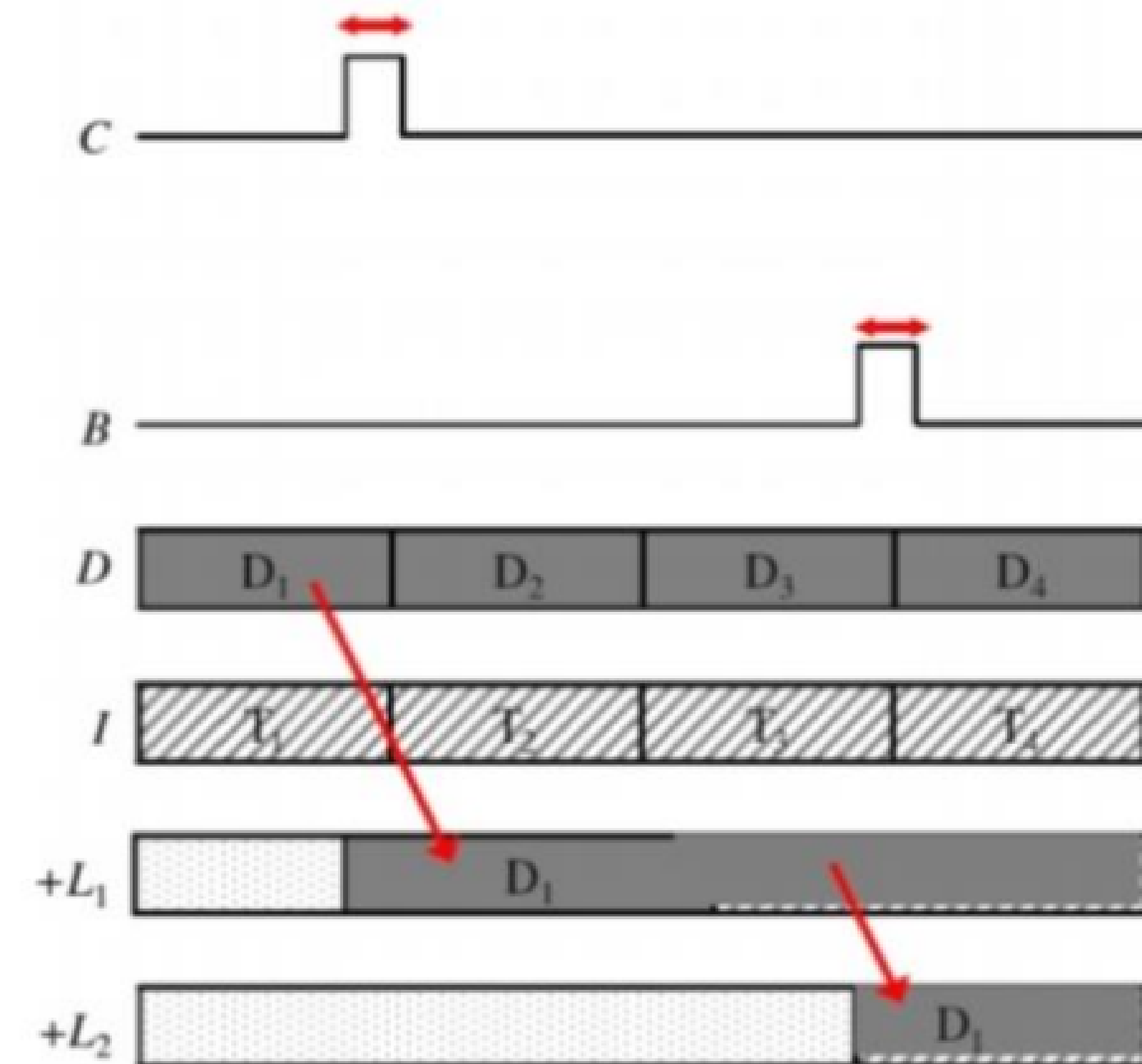
- **Clocking sequence:**

- C clock B clock: $D \rightarrow +L_1 \rightarrow +L_2$



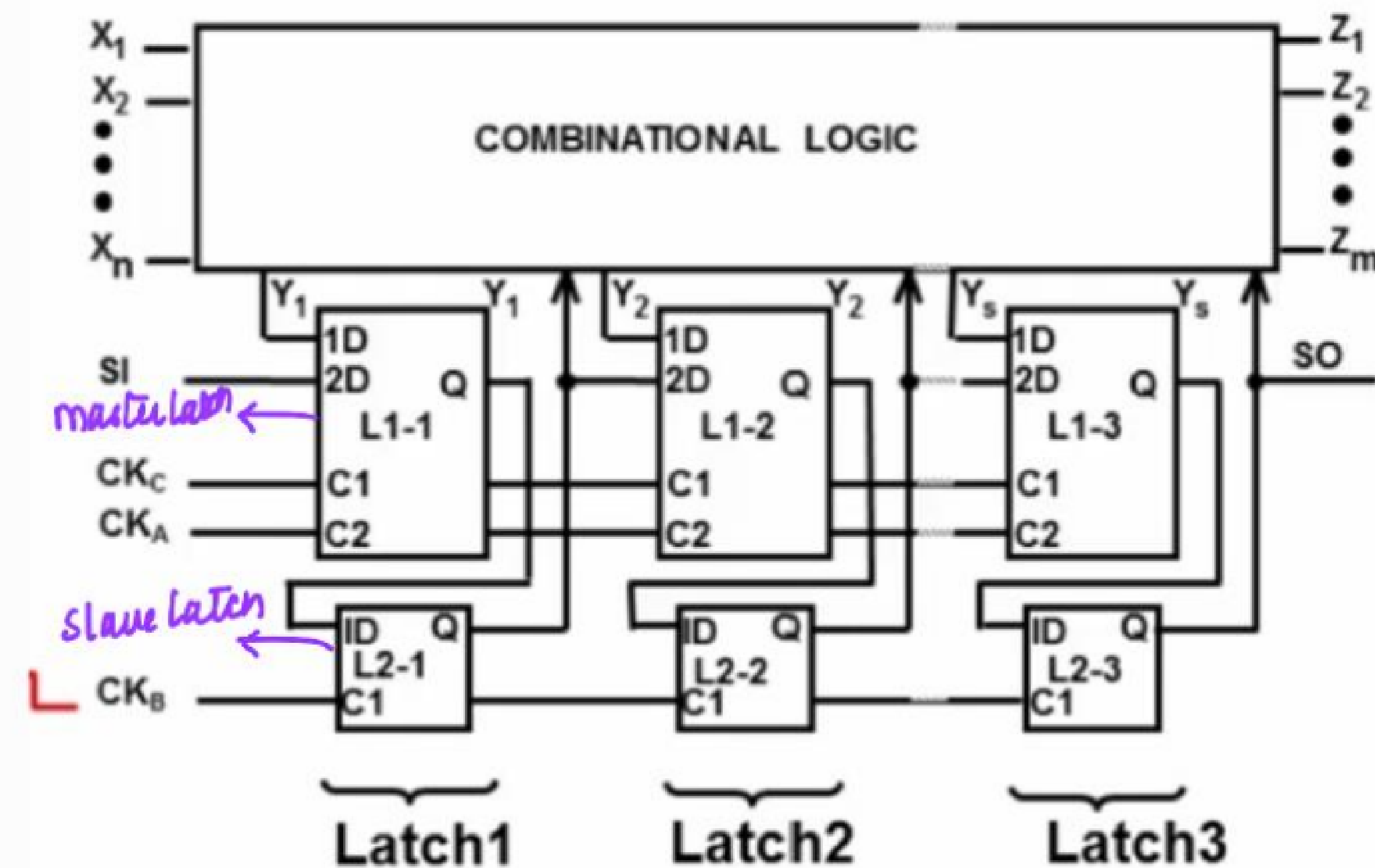
NAND TRUTH TABLE

A	B	y
0	0	1
0	1	1
1	0	1
1	1	0



Test Mode Operation (1)

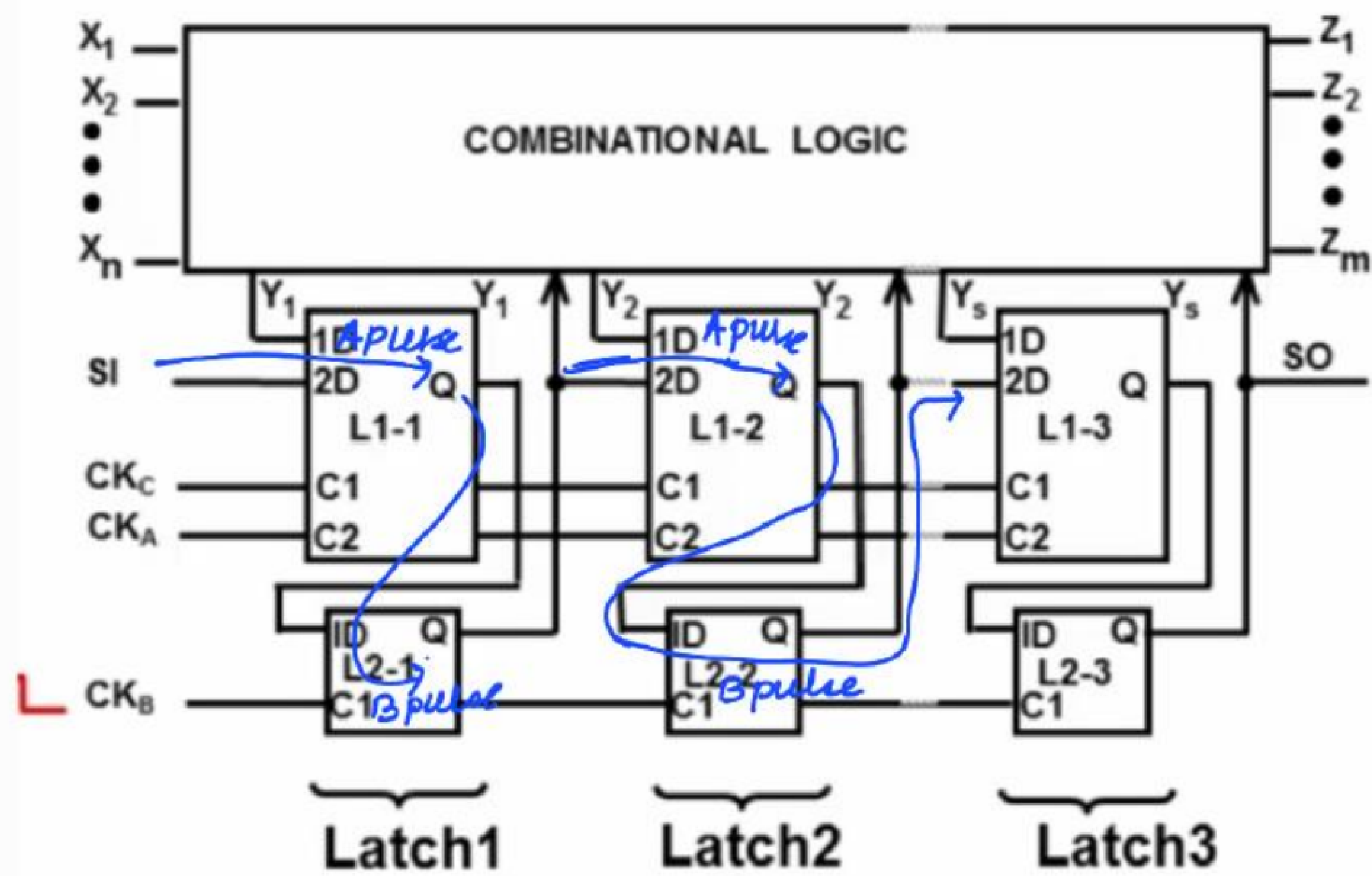
- One-pattern test
 - Shift: CK_A CK_B , CK_A CK_B , CK_A CK_B
 - Capture: CK_C
 - Shift: CK_B CK_A , CK_B CK_A , CK_B

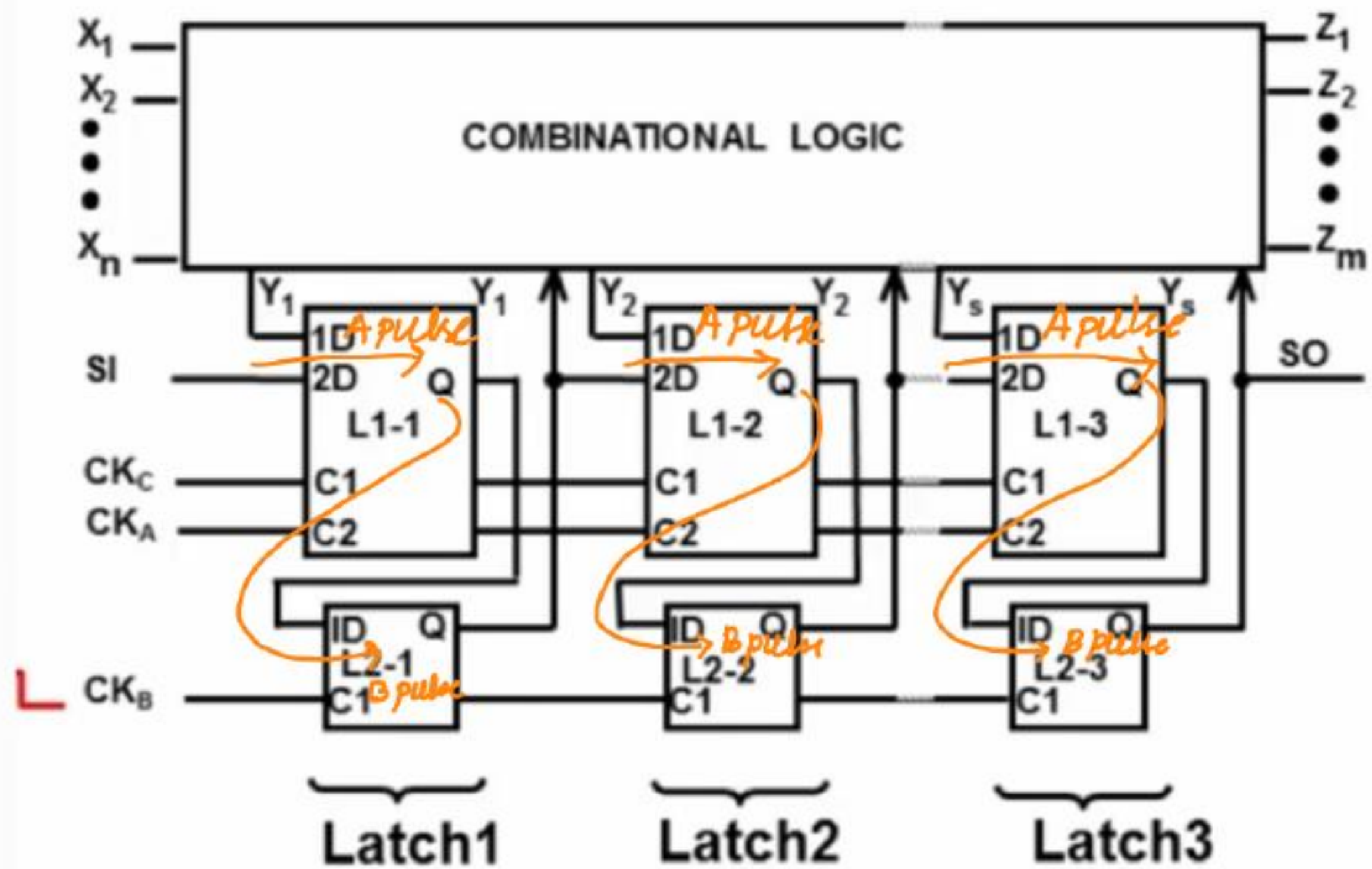


1st bit

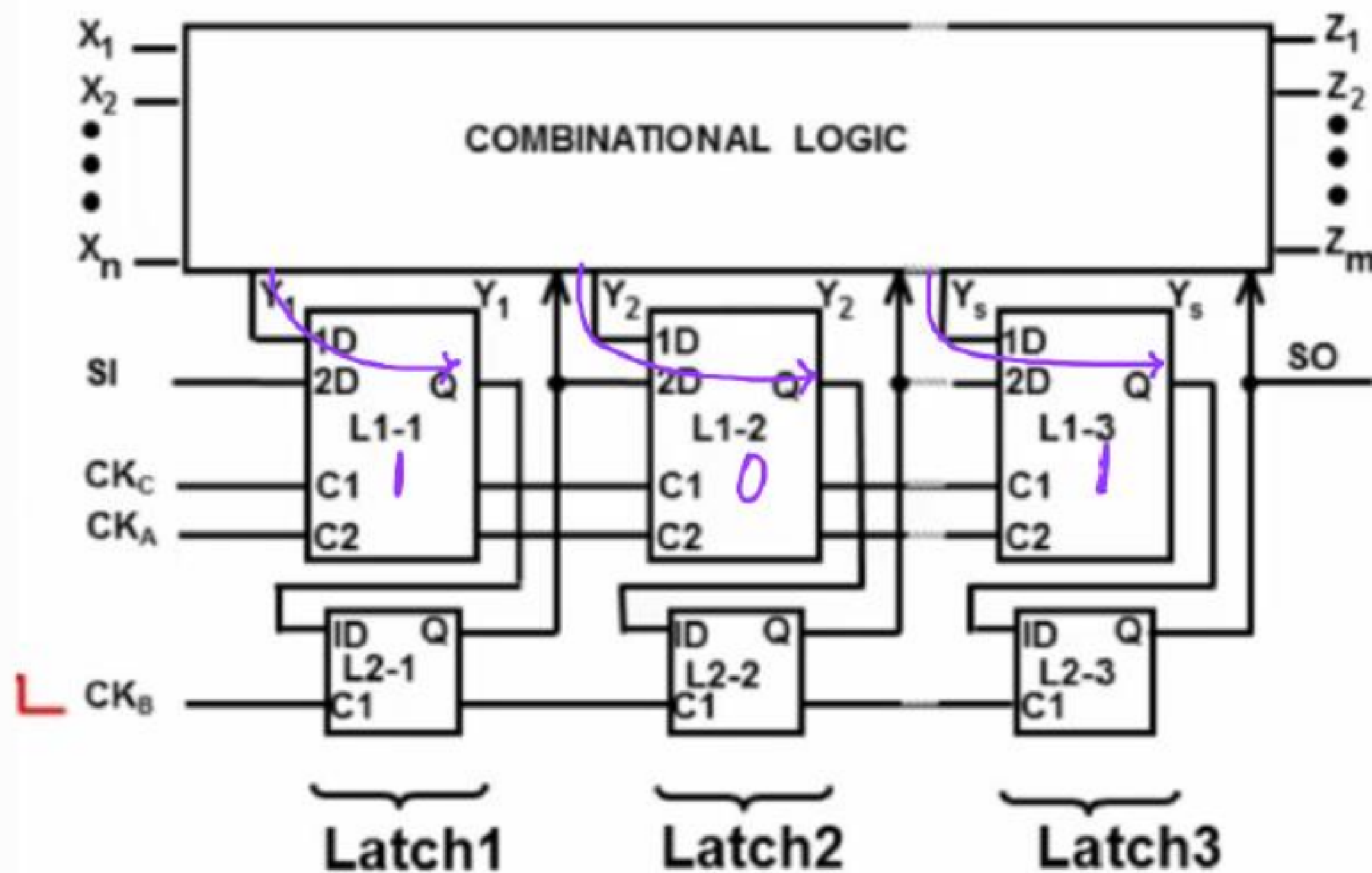
SI $\rightarrow$ +L1 $\rightarrow$ +L2





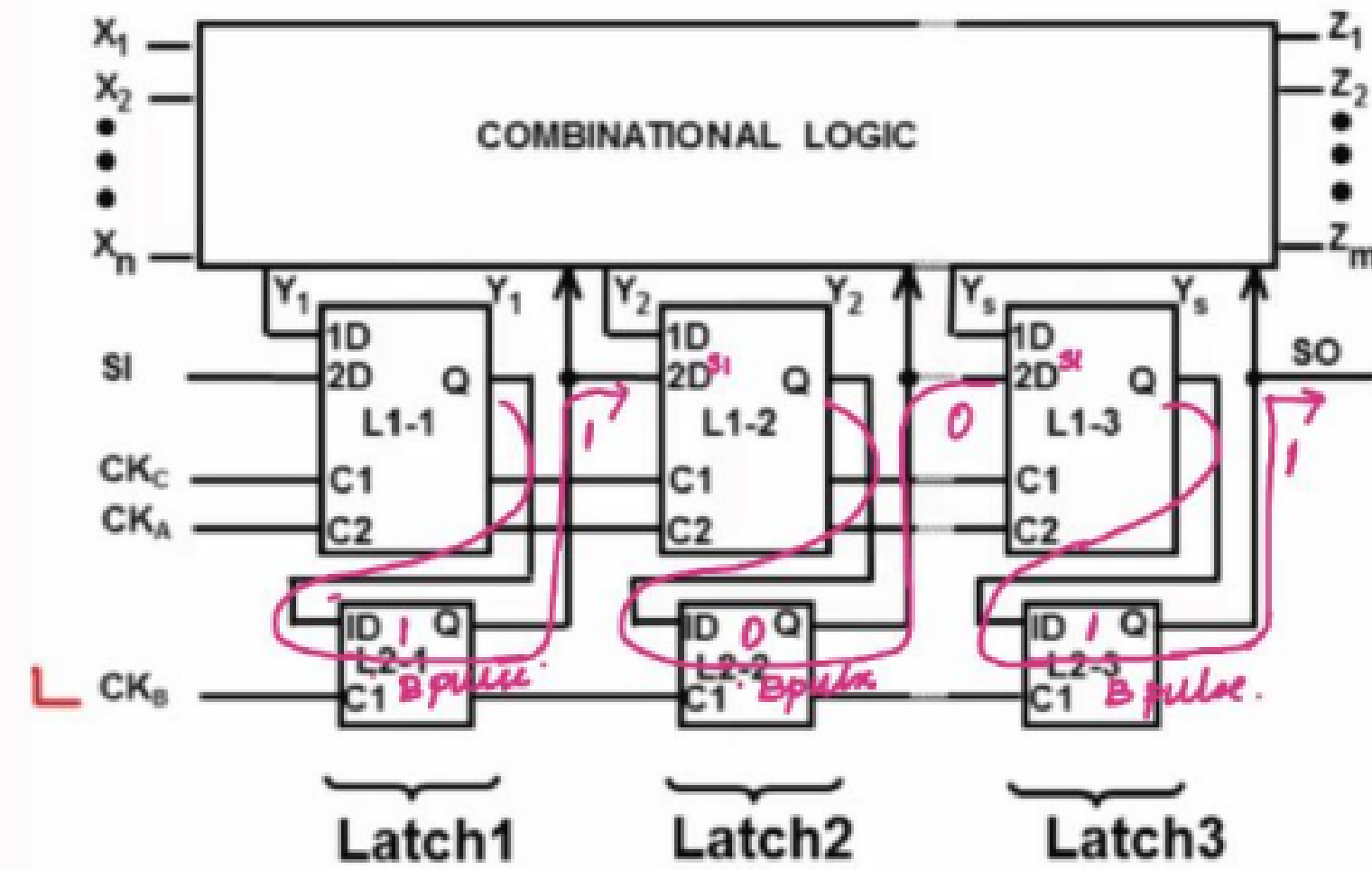


CAPTURE :- CK_C (Values shown in pic as just assumption)



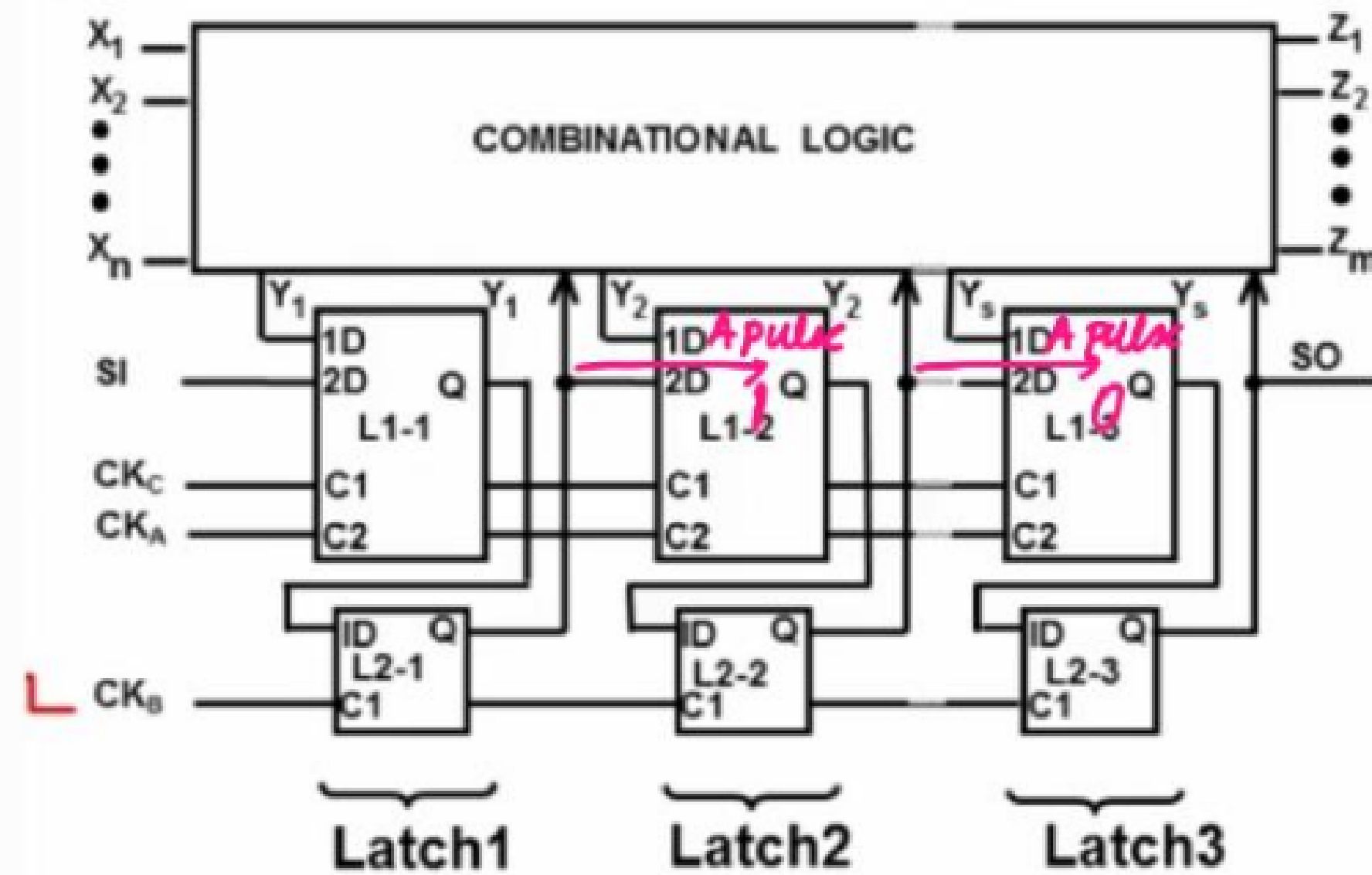
SHIFT :- $\rightarrow$ captured value gets shifted out

CK_B CK_A , CK_B CK_A , CK_B CK_A

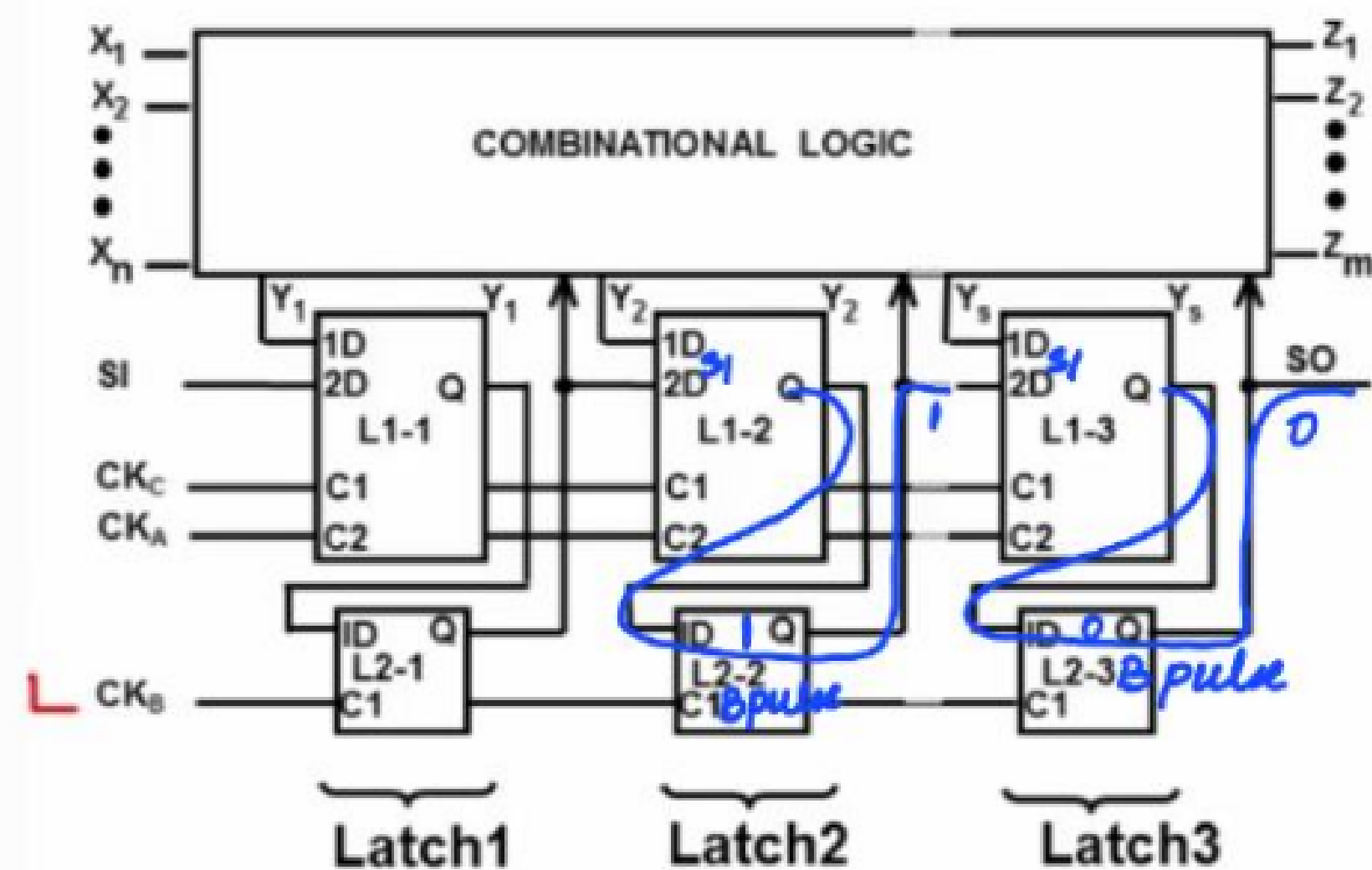


B pulse
 $+L1 \rightarrow +L2$

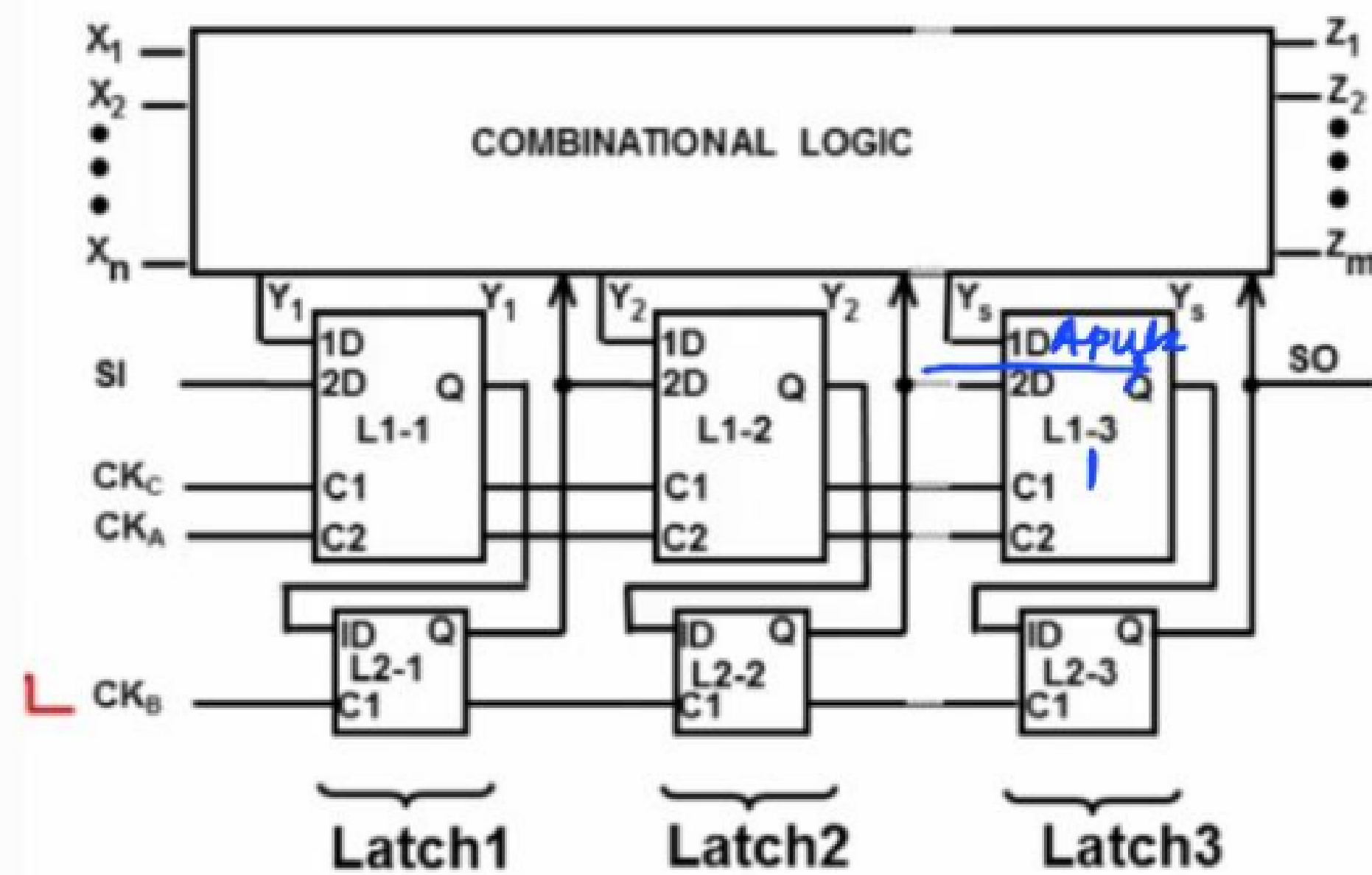
A pulse



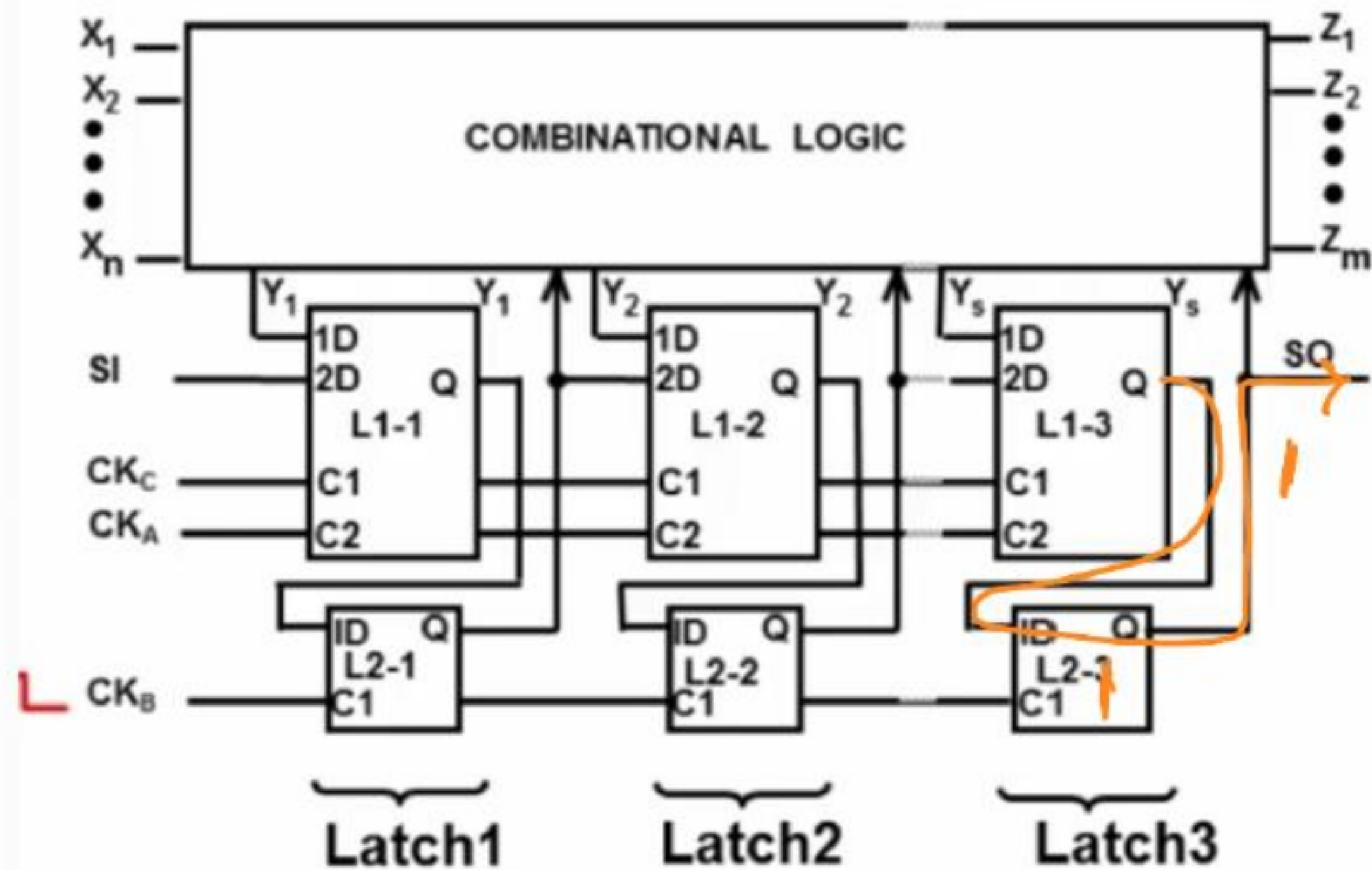
A pulse
 $SI \rightarrow +L1$



B pulse
 $+L1 \rightarrow +L2$



A pulse
 $SI \rightarrow +L1$



B pulse
+L1 $\rightarrow$ +L2

COMPARISION OF SCAN CELL DESIGNS

	Advantages	Disadvantages
Muxed-D Scan Cell	Compatibility to modern designs Comprehensive support provided by existing design automation tools	Add a multiplexer delay
Clocked-Scan Cell	No performance degradation	Require additional shift clock routing
LSSD Scan Cell	Insert scan into a latch-based design	Increase routing complexity